SM015
Mathematics
Semester 1
Session 2023/2024

MATHEMATICS(SM015)

SEMESTER 1

______________________SPECIMEN SM015___________________________

2 HOURS

1. This question booklet consists of 10 questions.
2. Answer **ALL QUESTIONS**
3. All steps must be shown clearly.
4. Numerical answers can be given in the form of π, e , surd ,fractions or up to three significant figures.

Differentiation and Integration :

$\frac{d}{dx}(\sin x) = cosx$ $\frac{d}{dx}(cosx) = -sinx$ $\frac{d}{dx}(\tan x) = sec^2 x$

$\frac{d}{dx}(\cot x) = -cosec^2 x$ $\frac{d}{dx}(\sec x) = \sec x \tan x$ $\frac{d}{dx}(cosec\ x) = -cosec\ x \cot x$

Sphere	:	$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$
Right circular cone	:	$V = \frac{1}{3}\pi r^2 h$	$S = \pi rs$
Right circular cylinder	:	$V = \pi r^2 h$	$S = 2\pi rh$

LIST OF MATHEMATICAL FORMULAE

Trigonometry :

$$\sin(A \pm B) = sinA\ cosB \pm cosA\ sinB$$

$$\cos(A \pm B) = cosA\ cosB \mp sinA\ sinB$$

$$\tan(A \pm B) = \frac{tanA \pm tanB}{1 \mp tanA\ tanB}$$

$$\sin A + \sin B = 2\ sin\frac{A+B}{2} cos\frac{A-B}{2}$$

$$\sin A - \sin B = 2\ cos\frac{A+B}{2} sin\frac{A-B}{2}$$

$$\cos A + \cos B = 2\ cos\frac{A+B}{2} cos\frac{A-B}{2}$$

$$\cos A - \cos B = -2\ sin\frac{A+B}{2} sin\frac{A-B}{2}$$

$$\sin 2A = 2\ sinA\ cosA$$

$$\cos 2A = cos^2A - sin^2A$$

$$= 2cos^2A - 1$$

$$= 1 - 2sin^2A$$

$$\tan 2A = \frac{2tanA}{1 - tan^2A}$$

$$sin^2A = \frac{1 - cos2A}{2}$$

$$cos^2A = \frac{1 + cos2A}{2}$$

Answer all questions

SECTION A

This section consists of 3 questions. Answer all questions.

1 (a) Evaluate $\lim_{x \to \infty} \dfrac{2x^2+8x-2}{x^2+1}$ [3 marks]

b) Determine the values of h and k for such that the following function is continuous at $x=-1$ and $x=1$

$$f(x)=\begin{cases} hx+k, & x \le -1 \\ 4x^2-5hx-k, & -1 < x \le 1 \\ -4 & x > 1 \end{cases}$$

[6 marks]

2 (a) Find the derivative of $f(x)=\dfrac{2}{3\sqrt{x}}$ using the first principles. [5 marks]

(b) It is given that $x^2y-\sin\left(2x^2\right)=0$ .Show that

$2y(1+8x^4)+4xy'+x^2y''-4\cos(2x^2)=0$. [6 marks]

3 Water is poured into vase at a constant rate of $80\pi\, cm^3s^{-1}$. When the depth of water is h cm, the volume of water, V cm^3 is given by $V=4\pi h(h+4)$. Find the rate of change of the depth of water in the vase when $h=6$ cm. [5 marks]

SECTION B

1 The complex numbers Z and W are given by $Z = 2 - i$ and $W = \dfrac{i}{1-i}$

Find $Z+W$ in the form of $a+bi$. Hence, express $Z+W$ in polar form.

[8 marks]

2 (a) Solve the inequality $\dfrac{x+1}{x-2} \geq 3$

[4 marks]

(b) Find the set of values of x for which $2 + \left|\dfrac{x}{4+x}\right| < 4$

[6 marks]

3 Given $p(x) = e^{-2x}$, $\mathrm{x} \in \mathrm{R}$ and $\mathrm{q(x)} = 1 + \mathrm{x}^3$, $\mathrm{x} \in \mathrm{R}$

(a) Find $p^{-1}(x)$

(b) Find $(q \circ p)(x)$.

(c) Solve $(p \circ q)(x)$ = e.

[7 marks]

4 (a) Function f is defined as follows :

$$f(x) = \begin{cases} e^x & , x \geq 0 \\ 1 - 9x^2 & , x < 0 \end{cases}$$

(i) Find $\lim_{x \to 0} f(x)$

(ii) Determine whether f is continuous at x = 0.Give reason for your answer.

[6 marks]

(b) Determine the value of p so that the function

$$f(x) = \begin{cases} 2x^2 + 8 & , x \leq 2 \\ 4px & , x > 2 \end{cases}$$ is continuous at $x = 2$

[5 marks]

5. (a) Find $\frac{dy}{dx}$ if $\mathrm{y}=\ln\sqrt{\frac{3+\mathrm{x}}{3-\mathrm{x}}}$. [5 marks]

(b) Given $x^2+4y^2-5xy=0$.Find y' when $x=1$

[7 marks]

6 Given that $x=\frac{1}{t+1}$ and $y=\frac{t^2}{t+1}$.

(a) Find $\frac{dy}{dx}$ in terms of t.

(b) Find $\frac{d^2y}{dx^2}$ at the point $\left(\frac{1}{2},\frac{1}{2}\right)$

[12 marks]

7. (a) Find the local minimum and the local maximum of the curve $y=x+\frac{9}{x}$.

[6 marks]

(b) A closed rectangular oil tank has a square base of length x m .

The total surface area of the oil tank is $96m^2$

(i) Show that the height of the oil tank is $\left(\frac{24}{x}-\frac{1}{2}x\right)m$.

(ii) Find the maximum volume of the oil tank.

[9 marks]

END OF QUESTIONS